

CUSTOMER SEGMENTATION WITH CLUSTERING

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problem overview

The e-commerce company operates on a large transnational dataset containing customer and order information across multiple continents. However, the business lacks a clear understanding of distinct customer groups and their purchasing behaviors. Without proper segmentation, marketing efforts may be inefficient and not tailored to different customer needs.

To address this, the objective of this analysis is to apply clustering techniques to segment customers based on key behavioral and value-based features, including frequency, recency, customer lifetime value (CLV), average unit cost, and age. The goal is to identify meaningful customer groups that can support targeted marketing strategies, improve customer retention, and increase overall business profitability.

Approach

The analysis followed a structured data science pipeline to perform customer segmentation. First, the dataset was explored and cleaned to address missing values, inconsistencies, and formatting issues. Feature engineering was then applied to derive key business metrics, including frequency, recency, customer lifetime value (CLV), average unit cost, and age.

As shown in figure 1, both Frequency and Recency exhibit right-skewed distributions, where most customers have low purchase frequency and relatively recent activity, while a small group displays significantly higher values, resulting in noticeable outliers. This highlights the uneven nature of customer behavior.

After preprocessing, the features were scaled to ensure equal contribution during clustering. Dimensionality reduction techniques such as PCA and t-SNE were used to visualize patterns in lower dimensions. K-Means and hierarchical clustering (via dendrogram analysis) were then applied to identify natural groupings, with the optimal number of clusters determined using silhouette score, inertia, and dendrogram structure.

Finally, the clusters were evaluated using boxplots and summary statistics to interpret behavioral differences and generate actionable business insights.

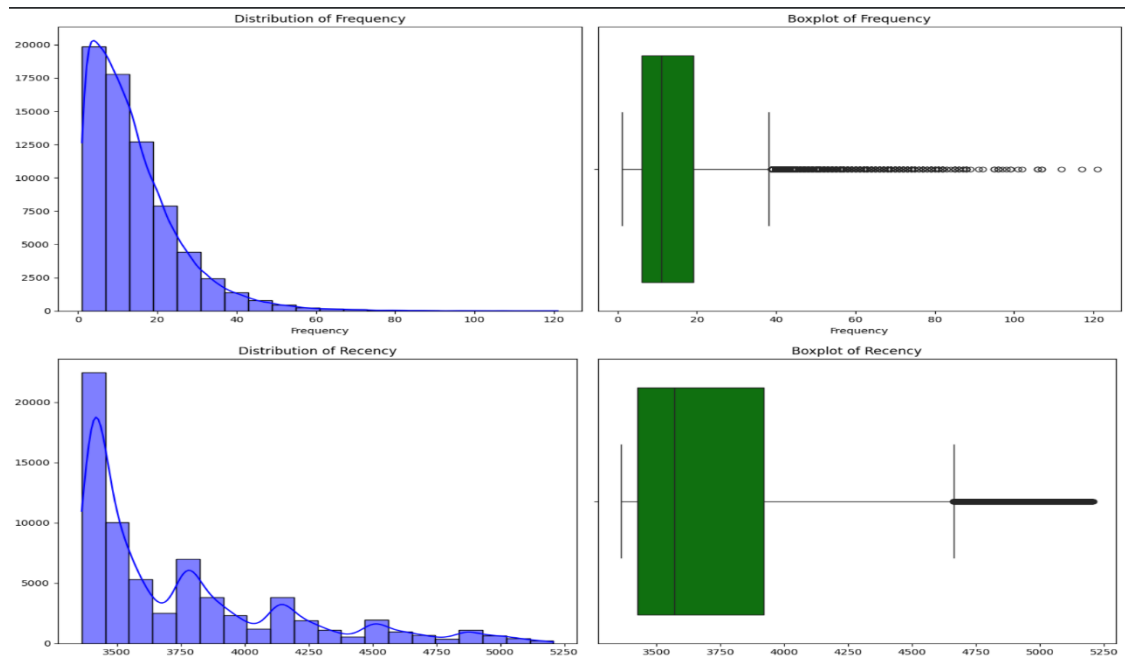


figure 1

Key Results

- IQR - Based Outlier Detection**

based on the result *figure 2* Using the IQR method, the dataset shows moderate variability with a small proportion of outliers across most features. These outliers likely represent natural variation in customer behavior rather than data errors and should be evaluated in the context of business meaning before removal.

```

Frequency outliers: 2654
Percentage: 3.8857979502196196

Recency outliers: 3353
Percentage: 4.909224011713031

CLV outliers: 2590
Percentage: 3.792093704245974

Age outliers: 0
Percentage: 0.0

UnitCost outliers: 2889
Percentage: 4.2298682284041

threshold 1 -> 13.76%
threshold 2 -> 3.05%
threshold 3 -> 0.01%
threshold 4 -> 0.00%
threshold 5 -> 0.00%
threshold 6 -> 0.00%

```

figure 2

- **Elbow Method**

The Elbow Method was used to determine the optimal number of clusters for K-Means by examining the within-cluster sum of squares (inertia). As shown in the *figure 3*, inertia decreases as the number of clusters increases, but the rate of decrease slows between 4 and 5 clusters. This “elbow” indicates the optimal number of clusters, where adding more clusters provides minimal improvement.

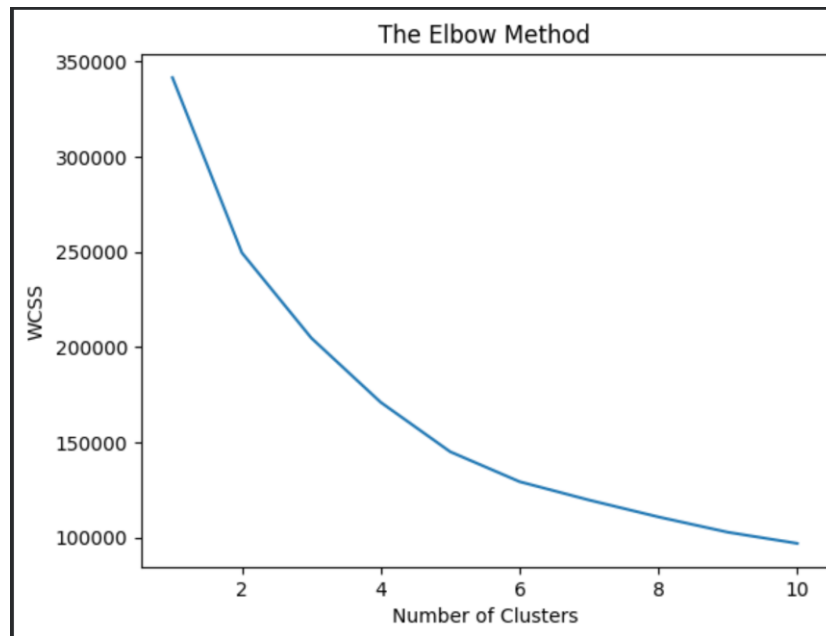


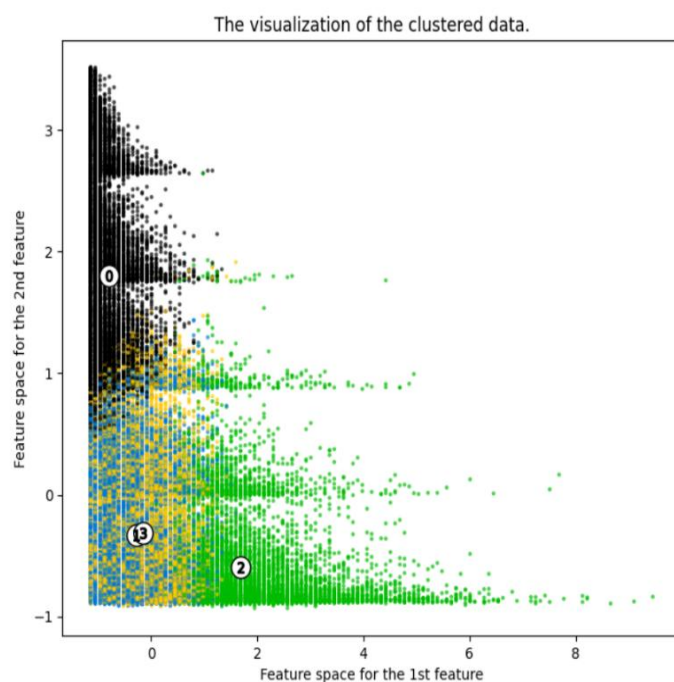
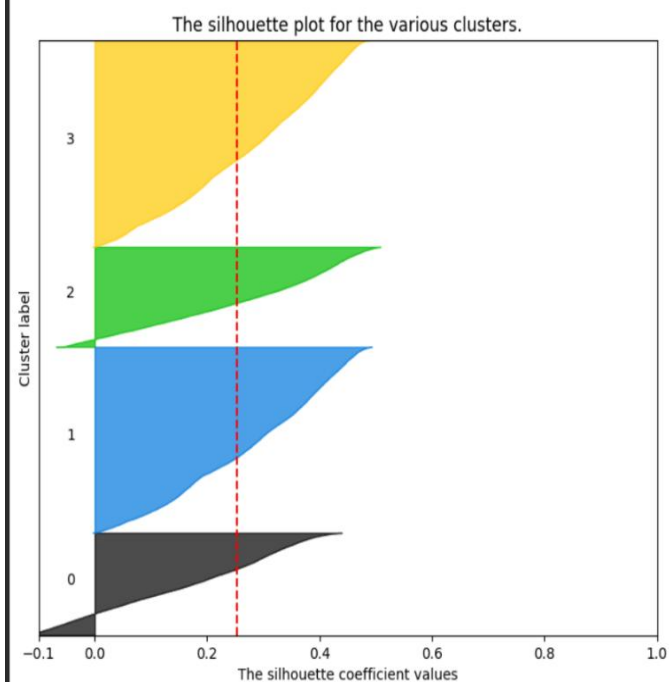
figure 3

- **Silhouette Score Method**

Silhouette method was used to evaluate the quality of the K-Means clustering by measuring cluster cohesion and separation. The results show that the 5-cluster solution achieved a slightly higher average silhouette score (0.267) compared to the 4-cluster solution (0.253), indicating marginally better-defined clusters as shown in *figure 4*. Therefore, the 5-cluster configuration was selected as the more suitable representation of the data.

For n_clusters = 4 The average silhouette_score is : 0.25316416819147647
 For n_clusters = 5 The average silhouette_score is : 0.266943479377642

Silhouette analysis for KMeans clustering on sample data with n_clusters = 4



Silhouette analysis for KMeans clustering on sample data with n_clusters = 5

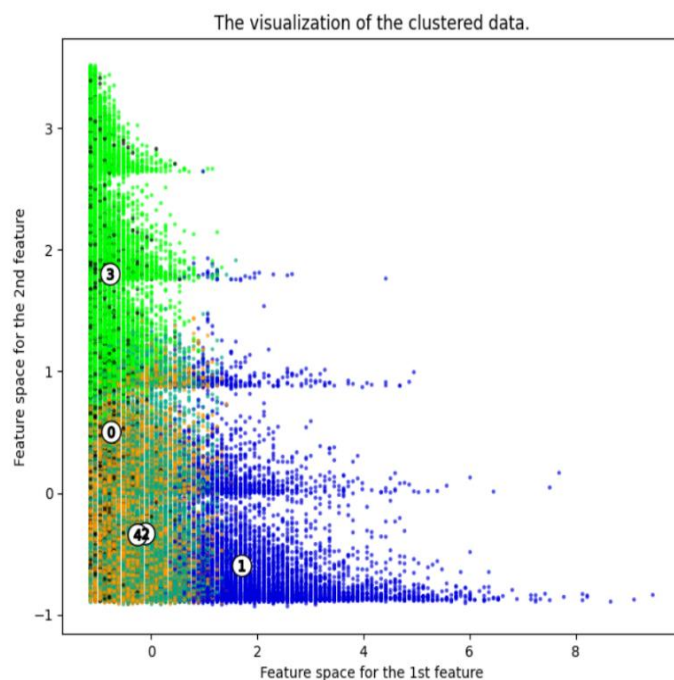
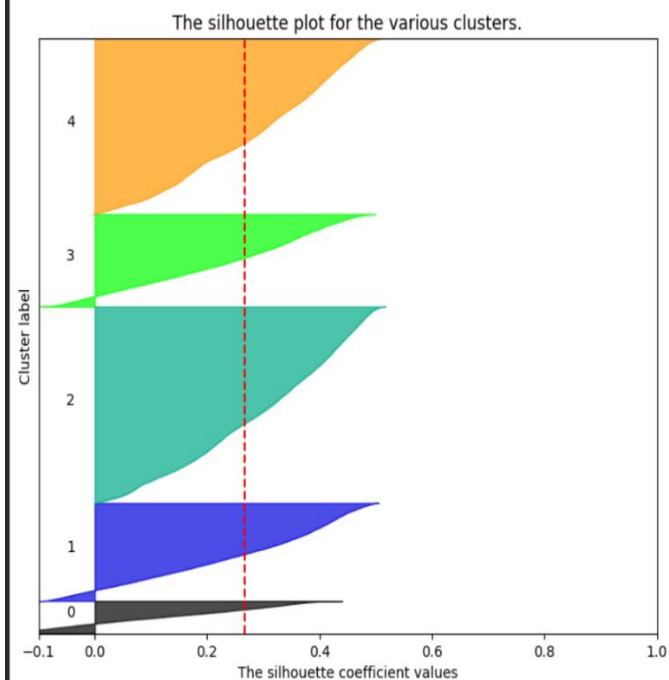


figure 4

- **Dendrogram (Hierarchical Clustering)**

The dendrogram was used to analyze the hierarchical structure of the data and identify natural groupings based on linkage distances. By observing the largest vertical gaps between mergers, a clear separation was identified, suggesting an optimal cut resulting in four clusters as shown in *figure 5*. This indicates that the data naturally forms four distinct groups based on similarity. The dendrogram results were used alongside the Elbow Method and Silhouette Score to support and validate the final clustering decision.

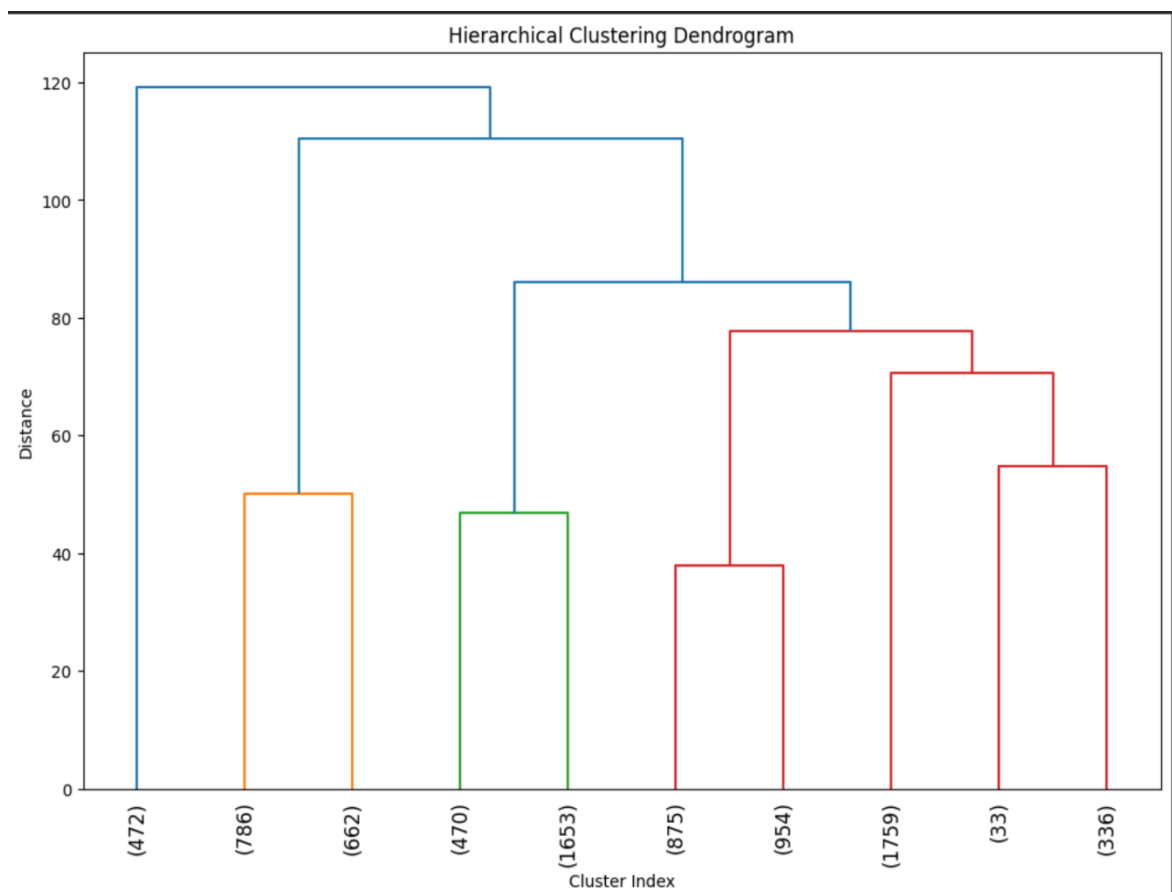


figure 5

- **K-Means**

```
k=4 inertia :171013.90748501313
```

```
k=5 inertia :145078.79444787616
```

K-Means clustering was applied to segment customers based on key behavioral and value-driven features. The algorithm was evaluated using different numbers of clusters, and performance was measured using inertia. The results showed that inertia decreased as the number of clusters increased, indicating improved cluster compactness with more groups. Specifically, inertia values were **171,013.91 for k = 4** and **145,078.79 for k = 5**, showing better within-cluster cohesion at higher k values. These results were further validated using silhouette scores and dendrogram analysis to ensure a balanced and meaningful segmentation.

	Customer ID	Frequency	Recency	Age	CLV	UnitCost	Cluster
0	1	35	3400	72.0	4322.70	89.622857	1
1	3	20	3502	43.0	2821.00	90.505000	3
2	4	9	4491	43.0	782.67	37.195556	2
3	5	19	3388	38.0	2810.98	107.947368	3
4	6	15	3573	67.0	1604.58	52.526667	0
...	...	...	...	...	...	...	...
68295	94248	6	4137	43.0	982.47	93.250000	2
68296	94250	4	4302	29.0	927.78	115.075000	2
68297	94252	49	3430	52.0	9311.27	77.312245	1
68298	94253	8	3726	58.0	1453.20	74.000000	0
68299	94254	6	4116	68.0	821.80	95.533333	2

68300 rows × 7 columns

- **Determination of Optimal Number of Clusters**

Multiple methods were used to identify the optimal number of customer clusters. The Elbow Method suggested the optimal number lies between 4 and 5, as adding more clusters beyond this range provided minimal improvement in inertia. The Silhouette Score peaked at $k = 5$, indicating better-defined clusters, while the dendrogram suggested four natural clusters based on hierarchical distances. Considering both statistical metrics and interpretability, the final solution balances cluster quality with actionable customer segmentation.

- **Cluster Distribution Analysis**

The boxplots summarize differences among the five clusters in terms of frequency and recency. Cluster 1 shows the highest median frequency with many high-value outliers, indicating highly active customers, while Cluster 2 has the highest median recency and greatest spread, reflecting varied engagement recency as shown in *figure 6*. These patterns highlight meaningful behavioral differences, supporting the use of these clusters for targeted segmentation.

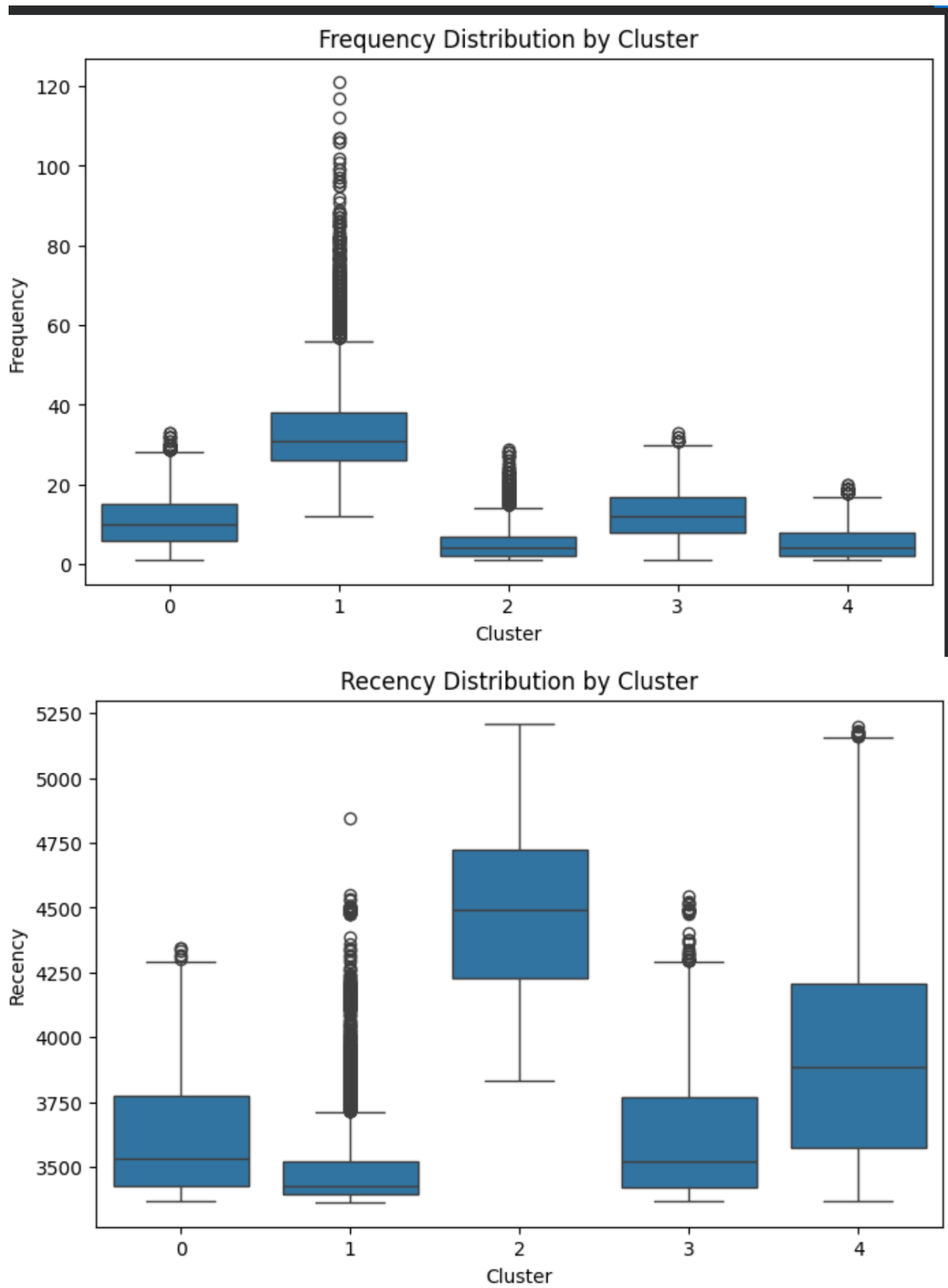


figure 6

- **PCA**

The PCA scatter plot on *figure 7* shows the five customer clusters in two dimensions, capturing the most variance with the first two principal components. Distinct, color-coded groupings confirm meaningful differences and validate the cluster assignments.

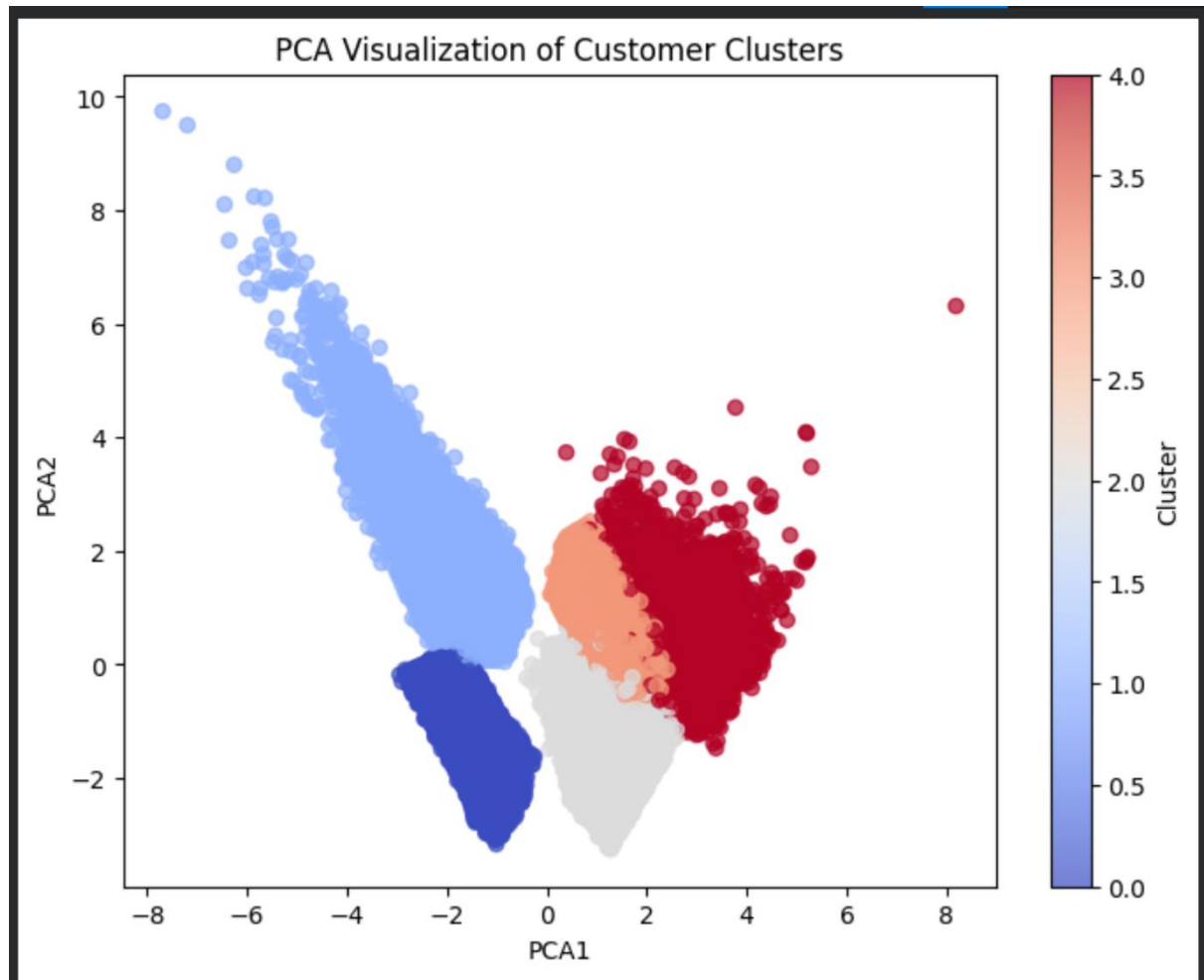


figure 7

- **T-SNE**

The t-SNE plot as shown in *figure 8* provides a two-dimensional nonlinear projection of the customer data, preserving local relationships to reveal the natural structure. The five clusters appear well-defined and visually distinct, confirming that the segmentation captures meaningful behavioral differences and validating the clustering results.

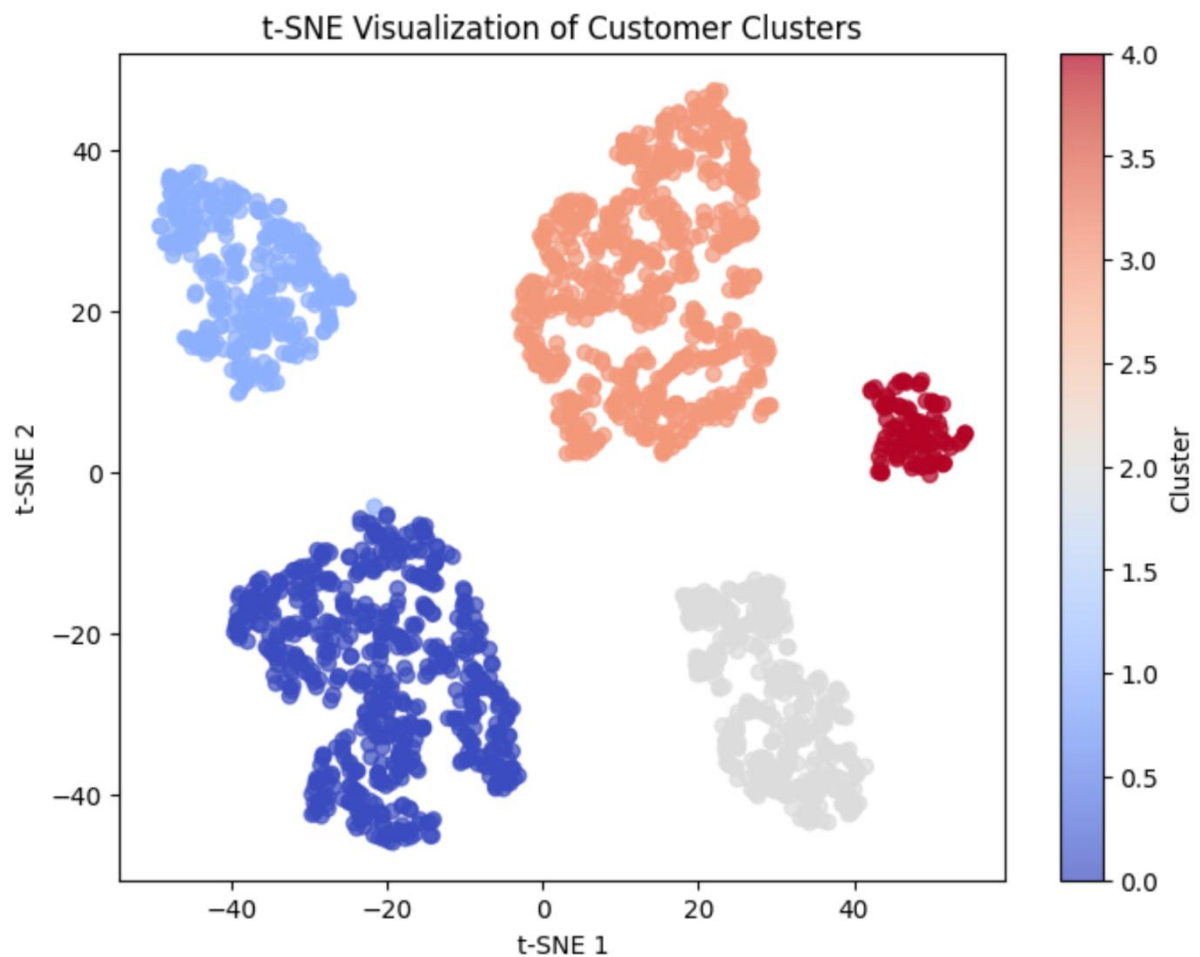


figure 8

Recommendations

Based on the customer segmentation results, the business should adopt targeted marketing strategies tailored to each customer group to improve retention and increase profitability.

High-value loyal customers should be prioritized through loyalty programs, exclusive offers, and personalized engagement to maintain their long-term value. Potential growth customers should be targeted at upselling and personalized promotions to increase their spending and engagement levels. At-risk customers require re-engagement strategies such as discounts, reminders, and targeted campaigns to prevent churn and restore activity. Low-value or occasional customers can be targeted with cost-effective marketing campaigns to encourage repeat purchases and increase frequency.

Overall, applying segment-specific strategies will enable the business to improve customer retention, optimize marketing resources, and maximize customer lifetime value.